

Adrian Diaz

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EDUCATION

California State University, Fullerton

Bachelor of Science in Computer Science

Fullerton, CA

Expected Graduation Date: May 2026

EXPERIENCE

Data Entry

April 2024 – Present

Smile Brands Inc.

Costa Mesa, CA

- Processed and configured dental insurance plans by entering structured procedure data and code specific fees into Dentrux Ascend, ensuring accurate system integration and compliance with office submitted tickets.
- Delivered prompt technical assistance to 600+ offices, diagnosing and resolving issues related to insurance configurations, fee schedule uploads, and data integrity.
- Implemented data validation protocols using advanced Excel functions to identify and resolve data inconsistencies, improving fee schedule accuracy by 25%.
- Automated recurring data transformation tasks with Excel macros, reducing manual processing time by 15%.

Software Engineer Intern

July 2021 – August 2021

University of California, Irvine

Irvine, CA

- Developed Python code for autonomous motion planning, modeling navigation logic and pathfinding algorithms.
- Optimized Dijkstra's algorithm to improve route planning efficiency for obstacle rich environments.
- Visualized algorithm behavior using Matplotlib and mathematical libraries to identify performance bottlenecks and enhance motion logic.
- Contributed to detailed technical documentation on algorithm design, facilitating team knowledge sharing.

PROJECTS

Accountabuddy | *React Native, Firebase, TypeScript*

January 2026 – Present

- Built a mobile application that increased user task completion rates by enabling social accountability through task management and sharing within a scalable social network.
- Engineered scalable social features allowing 100+ active users to add friends, share tasks, and track progress.
- Utilized Firebase to manage backend functionality, enabling task updates, notifications, and user interactions.
- Designed analytics and streak tracking to visualize user engagement trends and motivate sustained task completion over time.

Rubix Tutor | *Python, NumPy, Matplotlib*

August 2025 – January 2026

- Implemented an AI based Rubik's Cube solver using IDA* and IDDFS to search a large state space and generate optimal move sequences.
- Designed an admissible heuristic with a corner orientation pattern database, decreasing average solve depth.
- Modeled cube corner orientation and precomputed a pattern database with BFS to guide search.
- Developed a visualization interface with Matplotlib to display cube states and AI recommended moves.

Logfolio | *HTML, CSS, JavaScript, Node.js, Express.js, MongoDB*

June 2025 – August 2025

- Developed a full stack trade journaling web application to log, track, and analyze stock trades.
- Implemented secure JWT-based authentication and role-based access control for users and admins.
- Integrated external APIs (Finnhub, NewsAPI, YahooFinance) to provide market data, news, insider sentiment, and IPO events.
- Deployed the application using Render and MongoDB Atlas, ensuring scalability and reliability.

FindMyStuff | *HTML, PHP, CSS, JavaScript, MySQL*

January 2025 – May 2025

- Constructed a full stack web application to enable users to post, search, and manage lost and found item listings with image upload support.
- Developed secure server-side scripts to handle database CRUD operations and image processing.

TECHNICAL SKILLS

Languages: HTML, CSS, JavaScript, TypeScript, Python, PHP, C++, Bash, SQL

Frameworks / Libraries: Node.js, Express.js, React, React Native, NumPy, Matplotlib

Databases: Firebase, MySQL, MongoDB

Developer Tools: Git, GitHub, Postman, MongoDB Compass, Vercel, Visual Studio Code, Visual Studio, Linux